Can MLLMs Reason in Multimodality? EMMA Benchmark Overview

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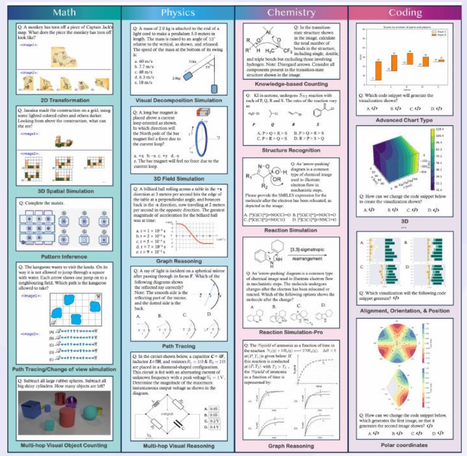
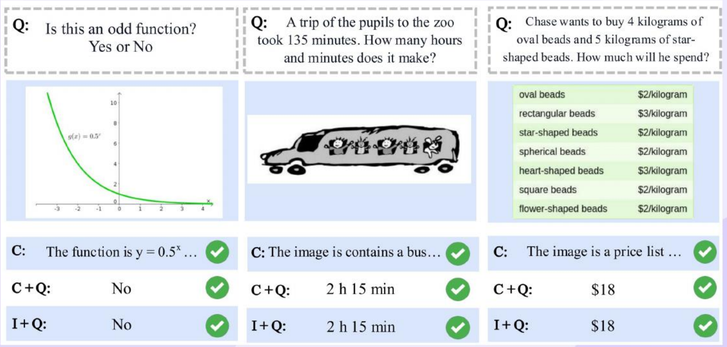
01 Why Multimodal Reasoning Matters



- **Human intelligence blends modalities**: text supports abstraction while images support spatial insight
- Real tasks demand integration, from design to scientific analysis, not siloed skills
- For MLLMs, coordinating visual interpretation with language reasoning yields more robust solutions
- Multimodal reasoning reduces brittle shortcuts and encourages stepwise, grounded problem solving

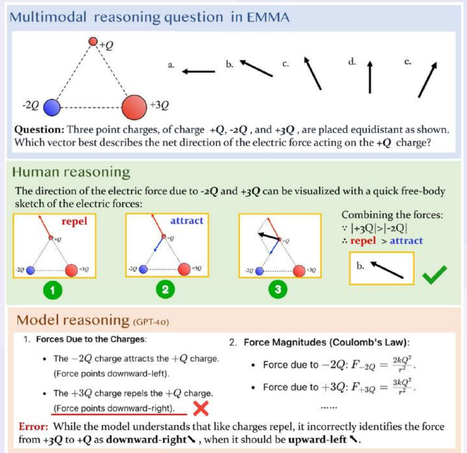
02 From Text-Only Progress to a Multimodal Gap



- LLMs show strong text reasoning; true multimodal reasoning remains unresolved
- Many benchmarks allow language-only shortcuts via captions, weakening visual tests
- We need items where visual reasoning is **essential**, not ancillary or replaceable
- Prior work highlights this leakage and calls for stricter curation (Lu et al., 2024b; Wang et al., 2024a; Zhang et al., 2024a)

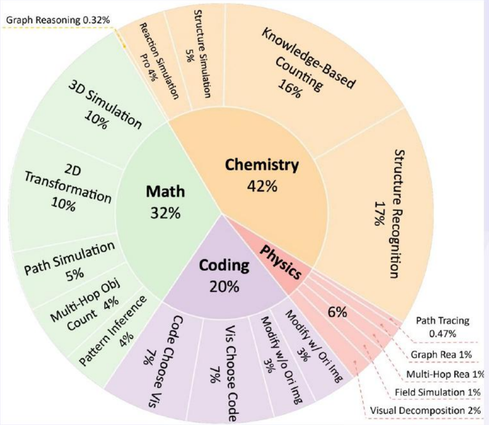
(Lu et al., 2024b; Wang et al., 2024a; Zhang et al., 2024a)

03 Introducing EMMA



- EMMA targets problems requiring both visual and textual reasoning steps
- Emphasizes back and forth interpretation across modalities, not one pass perception
- Avoids tasks solvable by text-only or a single glance, curbing shortcut solutions
- **Outcome**: clearer assessment of multimodal reasoning, not just recognition

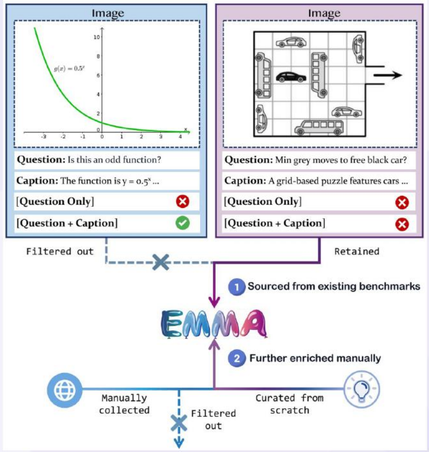
04 Domains and Skill Labels



- Four domains: **math**, **physics**, **chemistry**, **coding**
- Fine grained skill taxonomy enables targeted error analysis
- Two thousand seven hundred eighty eight questions with expert verified labels probe diverse multimodal skills
- Labels cover spatial simulation, graph reasoning, structure recognition, and visualization coding

05 Filtering Out Text-Only Shortcuts

- Two step filtering: text only, then text plus caption to remove leakage
- Discard any item if any model solves five or more of ten trials
- Ensures remaining items demand genuine visual reasoning beyond perception
- Stricter than prior pipelines, mitigating caption based shortcuts



(Yue et al., 2024a; Yue et al., 2024b)

06 Data, Human Baseline, Scope

Statistic	Number
Total questions	2,788
- Multiple-choice questions	2,002 (72%)
- Free-form questions	786 (28%)
- Questions with answers	2,788 (100%)
- Questions newly added	1,796 (64%)
Image in the question	2,599 (93%)
Image in the option(s)	195 (7%)
Problems with multiple images	298 (10%)

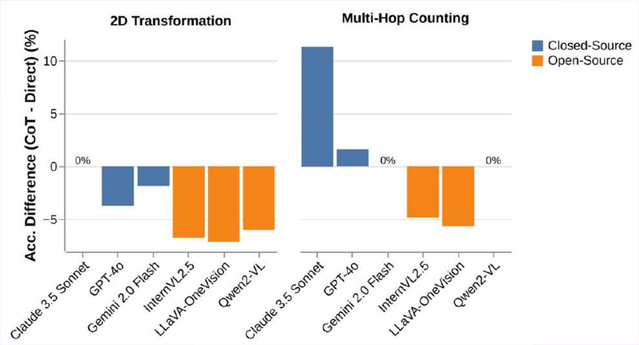
- EMMA mini: four hundred balanced questions across subjects
- Human experts provide baseline performance for context
- Short answer multiple choice and open ended evaluation with clear ground truth
- Zero shot evaluation across ten state of the art multimodal models

07 Headline Accuracy and Human Gap

	CoT	EMMA					EMMA-mini				
		Math (892)	Phys. (156)	Chem. (1,176)	Coding (564)	Overall (2,788)	Math (100)	Phys. (100)	Chem. (100)	Coding (100)	Overall (400)
Random choice	–	14.01	25.64	16.50	25.71	18.08	13.00	23.00	27.00	28.00	22.75
Human Expert	–	–	–	–	–	–	75.00	64.50	86.00	85.50	77.75
Claude 3.5 Sonnet	✗	25.34	33.97	40.90	38.65	35.08	23.00	34.00	44.00	35.00	34.00
Gemini 2.0 Flash	✗	23.88	38.46	36.31	42.02	33.61	20.00	40.00	36.00	41.00	34.25
GPT-4o	✗	27.24	38.46	31.89	40.07	32.42	30.00	38.00	33.00	40.00	35.25
Qwen2-VL-72B-Instruct	✗	33.07	42.31	32.06	34.57	33.46	38.00	40.00	34.00	37.00	37.25
LLaVA-Onevision-72B	✗	27.69	35.90	25.26	28.72	27.33	25.00	32.00	24.00	28.00	27.25
InternVL2-Llama3-76B	✗	25.11	22.44	24.06	27.84	25.07	31.00	22.00	21.00	28.00	25.50
InternVL2.5-78B	✗	31.39	38.46	35.20	31.91	33.50	30.00	40.00	38.00	33.00	35.25
Claude 3.5 Sonnet	✓	29.37	41.03	41.07	40.60	37.23 (↑ 2.15)	30.00	38.00	41.00	39.00	37.00 (↑ 3.00)
Gemini 2.0 Flash	✓	25.90	38.46	24.66	40.96	29.12 (↓ 4.48)	24.00	41.00	36.00	44.00	36.25 (↑ 2.00)
GPT-4o	✓	25.56	43.59	33.67	39.01	32.71 (↑ 0.29)	27.00	44.00	35.00	38.00	36.00 (↑ 0.75)
Qwen2-VL-72B-Instruct	✓	27.69	34.62	24.57	29.43	27.12 (↓ 6.35)	35.00	34.00	32.00	23.00	31.00 (↓ 6.25)
LLaVA-Onevision-72B	✓	22.42	15.38	22.70	30.67	23.82 (↓ 3.52)	23.00	26.00	23.00	29.00	25.25 (↓ 2.00)
InternVL2-Llama3-76B	✓	22.20	32.05	19.73	30.32	23.35 (↓ 1.72)	27.00	33.00	21.00	32.00	28.25 (↑ 2.75)
InternVL2.5-78B	✓	25.56	39.74	27.47	25.18	27.08 (↓ 6.42)	31.00	36.00	24.00	19.00	27.50 (↓ 7.75)
Gemini 2.0 Flash Thinking-1219	–	31.61	56.41	37.93	43.44	38.06	35.00	57.00	41.00	41.00	43.50
Gemini 2.0 Flash Thinking-0121	–	**37.11**	**60.26**	**41.58**	**48.05**	**42.50**	34.00	**63.00**	**47.00**	48.00	**48.00**
QVQ-72B-Preview	–	–	–	–	–	–	34.00	39.00	24.00	31.00	32.00
o1	–	–	–	–	–	–	**41.00**	49.00	40.00	**53.00**	45.75

- Best model on EMMA mini reaches forty eight percent; humans score seventy seven point seventy five percent
- Closed source models generally outperform open source across subjects
- Large gaps persist in **math**, **chemistry**, and **coding**
- Physics gap is smaller but still significant

08 Effect of Chain-of-Thought



- Chain of thought helps many closed source models but often hurts open source
- Task dependence: visual heavy tasks benefit less from textual chain of thought
- Open source models show more hallucinations with chain of thought
- **Implication**: language centric reasoning is not a substitute for visual simulation

09 Methods and Key Setup

Model	Method	Reward Model	N=1	N=2	N=4	N=8	N=16
GPT-4o	Majority Voting	–		–	37.25	36.25	38.25
	BoN	GPT-4o (Self)		35.50	35.75	36.75	–
	BoN	Gemini Flash Thinking	36.00	40.75	36.25	36.5	–
	Tournament	Gemini Flash Thinking		40.75	39.25	**41.25**	35.25
	Pass@N	–		45.00	53.25	65.75	74.00
Gemini 2.0 Flash	Majority Voting	–		–	37.75	39.25	39.75
	BoN	Gemini Flash (Self)		38.25	36.50	36.00	–
	BoN	Gemini Flash Thinking	36.25	36.75	37.00	40.25	–
	Tournament	Gemini Flash Thinking		36.75	37.25	**40.75**	38.75
	Pass@N	–		45.25	56.25	64.50	75.00
Gemini 2.0 Flash Thinking	Majority Voting	–		–	48.00	49.00	**50.75**
	Tournament	Gemini Flash Thinking (Self)	43.50	45.50	47.25	47.25	48.00
	Pass@N	–		53.75	64.50	71.50	81.50
o1	–	–	**45.75**	–	–	–	–

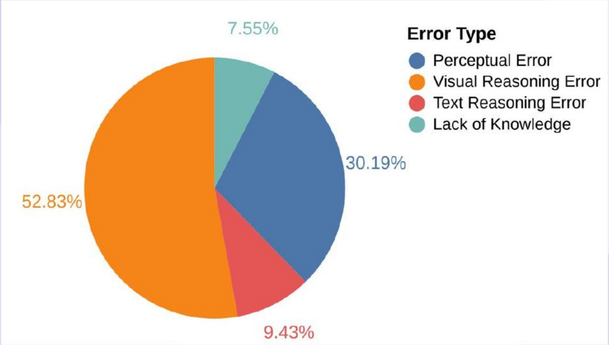
- Evaluate majority voting, best of N, and tournament selection
- Reward models: self versus stronger external for response selection
- Generate candidates with chain of thought; experiment with N equal to one, two, four, eight, sixteen
- **Goal**: quantify test time compute gains without retraining

10 Outcomes: Limited Gains

Model	Method	Reward Model	N=1	N=2	N=4	N=8	N=16
GPT-4o	Majority Voting	–		–	37.25	36.25	38.25
	BoN	GPT-4o (Self)		35.50	35.75	36.75	–
	BoN	Gemini Flash Thinking	36.00	40.75	36.25	36.5	–
	Tournament	Gemini Flash Thinking		40.75	39.25	**41.25**	35.25
	Pass@N	–		45.00	53.25	65.75	74.00
Gemini 2.0 Flash	Majority Voting	–		–	37.75	39.25	39.75
	BoN	Gemini Flash (Self)		38.25	36.50	36.00	–
	BoN	Gemini Flash Thinking	36.25	36.75	37.00	40.25	–
	Tournament	Gemini Flash Thinking		36.75	37.25	**40.75**	38.75
	Pass@N	–		45.25	56.25	64.50	75.00
Gemini 2.0 Flash Thinking	Majority Voting	–		–	48.00	49.00	**50.75**
	Tournament	Gemini Flash Thinking (Self)	43.50	45.50	47.25	47.25	48.00
	Pass@N	–		53.75	64.50	71.50	81.50
o1	–	–	**45.75**	–	–	–	–

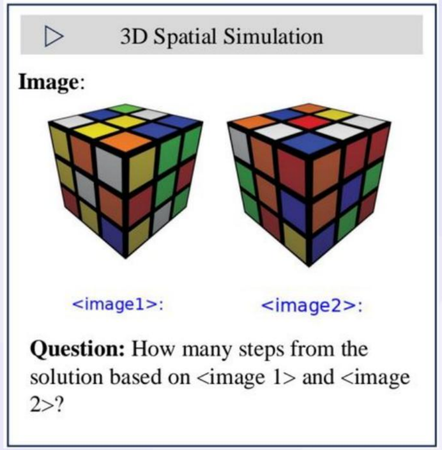
- Improvements up to about seven and a half percent yet far from human level
- Stronger reward models beat self reward; tournament exceeds best of N
- Benefits saturate for weaker bases; stronger bases raise pass at N
- **Takeaway**: scaling helps but does not close the multimodal reasoning gap

11 Underperformance and Visual Bottleneck



- Visual reasoning errors dominate overall failures
- Perceptual and textual errors are smaller fractions of mistakes
- Multi hop spatial simulation remains a recurring weakness
- **Implication**: better visual imagination and manipulation are needed

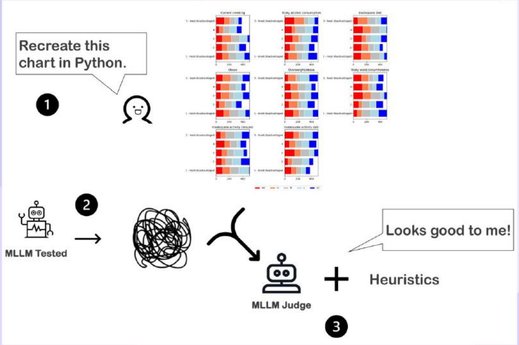
12 Relation to Prior Benchmarks



- EMMA removes caption based shortcuts to enforce visual reasoning
- Tasks are mapped to skill categories for cross dataset comparability
- Focus on problems unsolvable by text only or a single visual pass
- **Net effect**: stronger, more diagnostic multimodal assessment

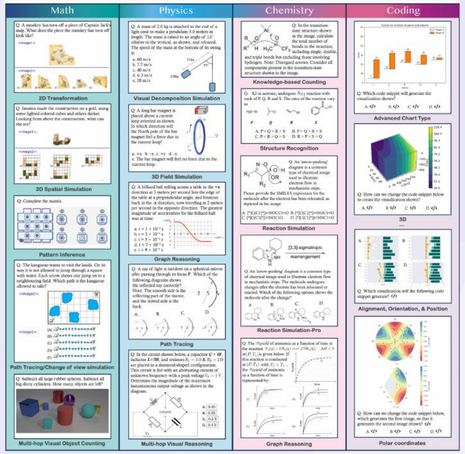
(Yue et al., 2024a)

13 Coding Task Design



- Multiple choice visualization tasks remove dependence on MLLM judges
- Fine grained categories enable targeted analysis of limitations
- Covers vision to code, code to vision, and modification workflows
- **Benefit**: reliable scoring and broader skill coverage

14 Conclusion and Impact



- A large human model gap persists on multimodal reasoning
- Visual reasoning remains the primary bottleneck across domains
- Progress needs new **architectures**, **training**, and **evaluation** paradigms
- EMMA provides a clear, diagnostics oriented benchmark for future work

https://emma-benchmark.github.io/

Thank You

Questions & Discussion